

Table 11: **Detection, repetition within responses, and diversity between responses.** Each method generates two 1,024-token responses for each of 50 prompts using Qwen3-8B-Base at temperature 1 with the full vocabulary. Self-BLEU measures average overlap in wording between the two responses to each prompt; lower values indicate greater between-response diversity. Repeated 4-grams is the percentage of contiguous four-token sequences that have already appeared earlier in the same response, counting every occurrence after the first. Distinct-3 is the number of distinct contiguous three-token sequences divided by the total number of three-token sequences in a response; higher values indicate less repetition. Both metrics include overlapping sequences and are averaged over the 100 responses. TPR is the percentage of watermarked responses detected. Each detector uses only the generated response and targets a false-positive rate of 0.1%. With repeat fallback on, SynthID, TextSeal, and Gumbel-Max use unwatermarked sampling at positions where the watermarking context repeats within a response. With it off, they continue applying the watermarking rule at those positions. Settings for the PRC watermark: online generation, $\eta = 0.05$, check weight $t = 3$, row rate 99/100, and posterior-mean/Hoeffding detection.

Method	TPR (%) $\uparrow$	Self-BLEU $\downarrow$	Repeated 4-grams (%) $\downarrow$	Distinct-3 $\uparrow$
Unwatermarked	–	0.01928	2.92	0.9514
PRC (ours)	97	0.01890	2.38	0.9561
Repeat fallback on				
SynthID depth 2	100	0.01987	3.28	0.9462
SynthID depth 10	100	0.02206	2.42	0.9554
SynthID depth 30	100	0.02890	2.30	0.9589
TextSeal $\alpha = .1$	100	0.04727	2.74	0.9464
Gumbel-Max	100	0.20718	2.81	0.9456
Repeat fallback off				
SynthID depth 10	100	0.02138	9.35	0.8880
TextSeal $\alpha = .1$	100	0.03770	32.17	0.6603
Gumbel-Max	100	1.00000	51.73	0.4720